Malcolm Steen

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https://steenmalcolm.github.io/malcolmsteen

Education

Georg-August Universität Göttingen Göttingen, Germany
M.Sc. Physics 04/2024 – Present
Current Grade: 1.2
Relevant Coursework: Advanced Statistical Physics, Introduction to Complex Systems, Statistical Physics of Disordered Systems

RWTH Aachen University Aachen, Germany
B.Sc. Physics 10/2019 – 10/2022
Final Grade: 1.5

Research Experience

Master Thesis 05/2024 – Present
David Zwicker Group, Max Planck Institute for Dynamics and Self-Organization

- Derive relationship between Flory-Huggins parameter and gradient terms from lattice model
- Find analytical approximation of surface tension in ternary system
- Develop a stepping-scheme algorithm to trace out coexistence curve in phase diagrams for N-component systems

Quantum Solutions Architect 11/2022 – Present
SavantX

- Develope a quantum approach for routing U.S. Air Force air cargo flights
- Designe hybrid quantum-classical algorithm that decomposes the problem into smaller subproblems suitable for current NISQ hardware
- Benchmark and compare quantum solution to state-of-the-art heuristics

Bachelor Thesis 04/2022 – 07/2022
Institute for Theoretical Particle Physics and Cosmology, RWTH University

- Implemented modifications to cosmological parameters in large-scale simulations and study the impact on structure formation
- Analyzed the power spectrum using mathematical models to identify new ways of incorporating beyond-Standard-Model physics into Newtonian simulations

Research Assistant: VISPA Project 10/2021 – 04/2023
Physics Institute III A, RWTH University

- Implemented SSH tunneling to allow secure access to compute nodes behind a gateway host
- Built a front-end UI for managing SSH keys and connections

Awards & Achievements

RWTH Dean's List(2022)
Awarded to the top 5% of students for outstanding academic performance

PhysicsGRE (2021)
Score of 930/990, placing in the 84th percentile

e-fellows Stipendium(2019)
Scholarship program that supports the top 10 students of each year

PSIA-AASI Certificate(2019)
Level 1 alpine ski instructor certificate

Skills

Programming: Python, C++, CUDA, JavaScript, React
Tools: PyTorch, JAX, Docker, Git
Scientific: Implicit/explicit/spectral numerical integrators, Bayesian inference, MCMC methods, SCFT, cavity and replica methods